

A Typology of Data Concepts and Contexts in Clinician-Patient Communication

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ABSTRACT

This paper presents a typology of concepts and contexts to inform the design of medical data visualizations for clinician-patient communication. As the accessibility, ubiquity and complexity of personal medical data grows, so does the need for patients to understand and interpret it correctly. Based on interviews with 19 healthcare professionals and analysis of current patient communication materials, our typology captures *i*) the diverse *types of medical data* communicated; *ii*) the *features* clinicians need patients to recognise and understand about these data; *iii*) *contextual information* required; *iv*) *motivation* for communicating data to patients; and *v*) current common *visualization techniques*. In doing so it identifies and defines the key dimensions of data-communication scenarios relevant to data visualizers working with patient data. We apply the typology to classify real world clinical scenarios and through doing so, illustrate how the typology can inform data visualization when designing communication aids for clinicians.

Index Terms: Data visualization, health communication, typologies

1 INTRODUCTION

Recent, rapid advances in medical technology have led to a huge increase in the volume and complexity of healthcare related data [1]. Combined with the growing accessibility of medical data [30] and move to empower patients through shared decision-making [28], patients are now exposed to data from sources such as continuous self-monitoring devices [17], electronic health records [14], and vital-signs monitors. In order to correctly retain and interpret this data, they must comprehend not only basic numerical concepts such as averages and percentages, but also higher-level concepts such as probability, variability, and error metrics [4]. Importantly, these must be understood in a medical context, then related to their own personal health. Specific examples of this include: identifying trends and thresholds when self-monitoring blood glucose values in diabetes [17] or understanding proportions in terms of relative and absolute risks when making treatment decisions [25]. This presents a challenge for clinicians: *how to communicate complicated medical data to patients in a way that is easy to understand?* [8]

Reasons patients struggle to correctly interpret medical data can include low levels of health numeracy or literacy [26], a heightened emotional state or poor communication with healthcare providers [4]. This can impact patient satisfaction and is associated with poorer health outcomes due to reduced treatment compliance and the inability to make informed medical decisions [25]. The communication of healthcare data hence presents a wicked problem [23, 8] for clinicians, which careful application of data visualization has the potential to alleviate. However, previous work in applying data visualization to aid patient communication of healthcare data has either focused on patients with long term conditions who self-monitor

and report their medical data to clinicians [22, 21] or specific scenarios, such as communicating personal healthcare data during in-patient stroke rehabilitation [24]. Here, we focus instead on a wide range of scenarios in which *clinicians explain data to patients*, independent of the specific health condition or clinical setting.

To that end, this paper presents a typology of concepts and conditions for the (visual) communication of data in healthcare. Our typology is constructed based on the following questions: *1) what is the medical data that needs to be communicated?* *2) what are the particular aspects, features or properties of the data that the clinician wishes to highlight?* *3) why do clinicians wish to highlight this data to patients?* *and 4) what is the contextual or background information required?* Based on these questions, our typology describes **data types** and **data features**, **contextual information**, **reasons for communication**, and **common visualization techniques** used by clinicians (Section 4). Medical data visualization for patient communication is unique due to an emotional, high-stakes context, the specific types of data and highly complex phenomena, the audience it caters to and the difference in practices and vocabularies.

Our typology is based on *a*) interviews with 19 clinicians and *b*) an analysis of 300 National Health Service (NHS) patient information sheets. We manually coded literature and expert interviews and iterated over our typology through discussion (Section 3). Finally, we demonstrate the descriptive power of our typology by describing two real world scenarios (Section 5).

2 RELATED WORK

2.1 Medical data and visualization

Systematic reviews of the medical literature have demonstrated that information visualizations can improve treatment-adherence in patients with chronic conditions [7] and aid the conceptualization of risk in cancer communication [29]. Studies in a variety of specific scenarios have demonstrated the benefits of visualizing medical data to patients or laypeople; e.g., aiding patient understanding of preoperative risks [6] or the communication of blood test results to patients with chronic kidney disease [12]. In parallel with this, recent work from the visualization community has focused on the application of specific techniques, for example narrative visualizations [18, 20], pictogram-timelines [15] or data comics [10, 31].

However, despite this success, existing work has generally been extremely scenario- or technique-specific and constrained to certain medical specialties. For example, whilst visualization has been used extensively to communicate data to patients with diabetes [20, 21, 9], there has been little work for respiratory conditions such as chronic obstructive pulmonary disease and asthma, despite the high prevalence and dependence on data, such as peak flow metrics, to inform treatment.

The lack of a universal schema for the application of data visualization techniques to medicine leads to missed opportunities for collaboration and discourse between the different fields. This is exemplified by the lack of diversity of data visualization practices applied to the current communication aids clinicians have at their disposal [2], and missed opportunities for medical data visualization. Our typology aims to address this gap by providing a comprehensive vocabulary and framework to discuss medical data

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and enable discourse and engagement between and within the medical and visualization communities. Furthermore, it allows for an assessment of the status of the field and identification of potential research opportunities.

2.2 Taxonomies in medicine

A recent series of articles on communicating numerical healthcare data by Ancker *et al.* (2025) [32] created a provisional taxonomy for data in healthcare, based on the communication of probabilities and numerical amounts to patients [2]. However, although this taxonomy includes the dimensions of *data-structures* and *data-presentation formats*, it is limited to only three high-level data-presentation formats, *words*, *numbers* and *graphics*, and does not provide a comprehensive typology of data visualization types. The work is also based only on a review of the research literature, rather than observations in current clinical practice. Other work from the medical community in patient data communication identifies the numerical concepts patients struggle to comprehend, such as uncertainty and risk [4]. Even though this can inform visualization design to some extent, important elements are missing, such as contextual information, and lower-level data concepts. Our work therefore complements this by adopting a data-centric view. It takes a granular and structured approach, to explicitly and comprehensively lay out the important aspects of medical data for visual communication. Importantly, it is based on clinicians' accounts to more accurately reflect current practice.

3 METHODS

Clinician interviews: The main body of the typology was constructed based on semi-structured interviews with 19 healthcare professionals. Recruitment was via mailing lists, attending networking at events, as well as leveraging professional and personal contacts. Interviewees included physicians (P1-P17), a research nurse (P18) and a dietitian (P19), working in patient-facing roles. Interviewees were from a wide range of medical specialties, including acute care, surgery, internal medicine, psychiatry and primary care, in order to obtain a comprehensive sense of the numerical medical data communicated, rather than specific challenges particular to one specialty. Experience ranged from newly trained (<5 years qualified, $n=4$), to moderately experienced (6-20 years qualified, $n=11$) and extensive experience (>20 years qualified, $n=4$). All were based in the UK, working in the NHS, as was the research nurse. The dietitian was based in Switzerland.

Core interview questions were the same for all interviewees. They covered the following areas: 1) common scenarios in which data was discussed with patients; 2) specific data discussed, including examples and framing; 3) objectives and goals of clinicians when communicating this data; 4) problems faced when presenting medical data to patients; and 5) tools, techniques and approaches currently used to explain medical data to patients.

We conducted semi-structured interviews in-person ($n=5$) or online ($n=14$), lasting 30-90 minutes each. All interviews were recorded and transcribed, then analyzed using a Template Analysis style of thematic analysis, as described by King *et al* [16]. We chose this style of analysis due to its highly structured nature, in which similar or related codes are grouped in a hierarchy, based around the key themes or dimensions. We iteratively revised the template of codes based on discussions among the authors.

Patient information sheets: To validate our typology, we downloaded the full library of approx. 1,200 currently used, reviewed and authorised patient information sheets from the Oxford Universities Hospital website [19]. These generic patient information sheets cover all medical specialties, including those not accounted for in our interview candidates. Patients are given them for a variety of reasons such as to provide information about procedures,

give instructions or aid in decision making and thus they often include medical data. We then manually analysed a random sample of 300 of these sheets by applying our typology. We aimed to identify our typologies concepts' such as data types, data features, data visualization and communicative intent. An illustrated example of this can be found in the supplementary materials, whilst Section 5, Figure 1 shows the application of our typology to a real world clinical scenario). We could entirely classify all the sheets and did not need to introduce new concepts to our typology. This shows that the typology is representative of the current domain of medical data communicated to patients.

4 FINDINGS

This section details the five dimensions of our typology, three dimensions capturing *What* is communicated; two dimensions capturing *Why* and *How* it is communicated, respectively. A detailed table with individual quotes from our interviews for each concept described can be found in the supplementary materials.

4.1 WHAT: Type of Data (Type)

The **type** of data being communicated is fundamental to the design of any data visualization. While many taxonomies for data types exist e.g., [27], here, we relate the data types to the specific medical data being communicated. Our interviews and the literature review indicated that personal medical data comes from a variety of sources, in a variety of formats defined and illustrated here. A wide variety of **Images or scans**, e.g., X-ray, magnetic resonance imaging (MRI) scan or photographs are used by clinicians when diagnosing and monitoring conditions. **Nominal values**, e.g., blood type, sex and **Ordinal values**, e.g., categories of obesity, disease stages, are often used to classify subgroups of patients. **Continuous Numerical values** may comprise **series** of data such as time series used to monitor conditions e.g., continuous blood glucose levels, electrocardiograms (ECG) or **individual values**, such as manually or machine-recorded values: e.g., pulse rate, blood pressure, blood test results. **Area or volume**, e.g., regions of rash or redness, volume of urine produced, often communicated as a continuous numerical value (e.g., percentage total body area of a burn, volume in ml of urine produced). **Scores** calculated based on multiple values from different measurements as opposed to one measured value e.g., Glasgow Coma Scale (GCS) Score are often used when assessing risk or triaging (prioritizing) patients for treatment. **Text notes** e.g., in medical notes and patient health records and finally **Video**, dynamic data from procedures e.g., footage from endoscopies or colonoscopies are also common.

4.2 WHAT: Data Feature (Feature)

The interpretation of these medical data often relies on important 'features' or 'observations' that are relevant to the message of the communication. Thus, the **data feature observed** dimension captures the specific observation or property of interest in a data set, e.g., the probability of x occurring, a trend or an outlier. When explaining data to the patient, clinicians need to emphasise these features in the data, depending on the goals of the exchange [3]. For example, a clinician working with a long-term patient to achieve weight loss may wish to emphasize overall **trends** in the data, rather than smaller **fluctuations** that might dishearten the patient. Conversely, a clinician working with a newly diagnosed diabetic may wish to highlight the danger of fluctuations in blood sugar levels. We now describe the different features clinicians refer to, and highlight the importance of distinguishing between related concepts such as probabilities and proportions.

PROBABILITY VS PROPORTION: A frequent data feature communicated is the predicted **probability** of an event occurring for that patient, or the measured **proportion** of patients it has previously been observed in. This is usually associated with explaining

risks and benefits. For example, when obtaining informed consent for a surgical procedure a *known complication rate* may be communicated as a percentage, or fraction, and used to estimate the *predicted risk* for that individual. Clinicians emphasize the necessity to distinguish between probability (predicted) and proportion (known) data due to the uncertainty associated with predicted values and ethicolegal guidelines they must follow. Probability can also be expressed as a function of time, e.g. when explaining a prognosis to a patient following diagnosis.

THRESHOLDS AND OUTLIERS: **Thresholds** are commonly used in evidence based medicine. They often refer to cut-off bands e.g., overweight, type 1, type 2, type 3 obesity based on BMI. This may also be used to indicate when a treatment or referral should be initiated. For example, clinicians may have to explain to patients they can only be referred for bariatric surgery if their *BMI is greater than 40*. A related feature is an **outlier**. These are extreme values that can indicate error or cause for alarm, e.g., extreme blood test result values due to contaminated samples.

MISSING DATA: Medical data is full of gaps. For example, medical data for certain patient populations, such as the elderly or those with multiple diseases is often missing and this must be acknowledged when discussing do not resuscitate (DNR) orders and end of life care.

RELATIVE VS ABSOLUTE CHANGE: **Relative** and **absolute change** are extremely important to distinguish between in health-care data. This is particularly true when communicating notions such as the benefits or risks of treatment, due to small changes in **absolute risk** that correspond to huge changes in **relative risk**, something acknowledged by the medical community as problematic [4]. However, it is also true in other, less recognized situations (see Section 5: Example Scenario 1; Diabetes Review Appointment).

FLUCTUATIONS OR TRENDS: Clinicians highlight the difference between **fluctuations** and **trends** in a series of data, with the former referring to transient aberrations in the data e.g., blood sugar levels spiking after meals in poorly controlled diabetes and the latter referring to the long term direction of the data set, such as a gradual deterioration in kidney function over time.

FREQUENCY: The **frequencies** of events are common when evaluating patients (e.g., number of heartbeats per minute (heart rate)) or instructing them (e.g., the number of times to take a medication each day). It is essential to emphasize the units and regularity of frequencies, e.g., when prescribing medications

RANKING: We use the term **ranking**, to refer to mapping data points in relative order of severity or importance. For example, the stages of cancer map the degree of spread (early, localised, regional, distant) to a number (1-4) whilst the degree of psychiatric symptoms map emotional state to a Likert Scale. Alternatively, clinicians often have to emphasize the relative importance of taking a medication.

INTERACTIONS: Medical data is increasingly multi-dimensional, for example patients may have multiple conditions or symptoms and be on a number of medications (polypharmacy). This is further complicated if the different dimensions influence each other. For example, patients with type 2 diabetes often have kidney disease. Metformin, a preferred treatment for type 2 diabetes can reduce kidney function. Therefore, kidney function must be carefully monitored as the metformin dose is increased until an optimal dosage is achieved, balancing the benefit of better diabetes control against the risk of kidney failure.

INDICATORS: Disease is monitored using indirect measurements, termed markers or indicators. For example, white cell count or C-reactive protein levels are measured in the blood to monitor infection levels. Sometimes these indicators can seem obscure to a patient and require clarification, for example an increased HbA1c level in a blood test reveals poor long term diabetes control.

4.3 WHAT: Contextual Information (Context)

It often is necessary to include certain contextual information when communicating medical data, as patients often do not understand anatomy, body systems or medical equipment. Thus, patients can misinterpret instructions or misunderstand procedures.

ANATOMY: Unsurprisingly, **Anatomy** is the most common piece of contextual information. Patients' understanding of it varies enormously, and can cause misunderstandings: *"So the classic thing with a vaginal birth after caesarean section is a risk of scar rupture [...] everyone automatically thinks I mean the scar on their abdomen, the scar that they can visibly see [...] I have to be explicit that I am talking about the scar on the uterus"* [P14].

EQUIPMENT AND PROCEDURES: Patients must also be familiar with the medical **equipment and procedures** used e.g., to administer medication. Failure to use equipment correctly can cause severe consequences [14]. Patients are often curious about the equipment clinicians use during procedures stimulating discussion around medical data: *"...there's often generic questions like 'why so many tubes [and] colours'; then I think that's about explaining to them the different types of tests and why."* [P2].

PHYSIOLOGY: Finally, clinicians refer to the need to explain **physiology**: bodily processes monitored or indicated with data. For example, the course of brain death, a predictable cascade of events that occurs before a person is removed from life support: *"coning: that's basically when the intracranial pressure rises, the cerebral hemispheres swell and put pressure on the brain stem so no blood gets in ... the brain stem stops working and then dies... once somebody's got fixed dilated pupils, they're dying because of coning..."* [P03].

4.4 WHY: Communicative Intent (Intent)

It is vital to define the intent of communication when explaining medical data to patients in order to decide which data and features to highlight. This dimension addresses questions such as: does the clinician wish to evaluate, instruct, objectively support decision making or give future predictions? The goal of communication is highly relevant to data visualization, e.g., in health risk communication graphical features that improve the *accuracy* of quantitative reasoning when decision making, differ from features most likely to *alter patient behavior* when emphasising the importance of treatment adherence [13].

DECISION SUPPORT: Decision making requires presenting patients with treatment options, ensuring appreciation of the **risks and benefits** for each. It is essential in modern medicine due to shared decision making [28] and informed consent [5]. In the context of our typology, we constrain this to patient-led decisions as opposed to those made exclusively by the medical team. If the patient has no input into the decision, the goal of communication becomes **instructing**. Decision making also covers the process of obtaining informed consent. In terms of work to date, it is the most studied area in medical visualization.

EVALUATION AND EXPLANATION: Evaluation refers to analysing and explaining the patient's existing medical data. It involves the recognition of previous **trends, fluctuations** or **outliers** and relating these to the patient's experience. In particular, it focuses on the current health status of the patient, relating it to past known values, but *not* on future predictions. As this typology focuses specifically on data communicated by the clinician to the patient, evaluation does not encompass self-monitoring data from the patient, for which other research exists [22]. Evaluation is a key component of communication when a patient presents in a clinic for follow-up or when making a diagnosis.

INSTRUCTION: Clinicians must give patients clear directions in how to take medications or respond should their health status change (e.g., 'safety-netting' [11]). We define situations as **instructive** if a clinician conveys information, with the aim of promoting

recognition or recall, rather than decision making. For example, remembering what requires self medicating versus what constitutes an emergency when taking medication.

PREDICTION: We define **prediction**, as a communication in which a patient's prognosis or outcome is forecast. Prediction is distinct from other communicative intents, in that it is speculative, based on medical evidence but plagued by uncertainty and this must be emphasised. It is one of the most challenging communication tasks clinicians face, due to its often high-stakes and emotive nature. Predictive visualizations may include illness trajectories, mortality rates and life expectancies or *probabilities* that an event might occur for that patient (as opposed to the proportion of patients it previously occurred in).

4.5 HOW: Current Data Visualization Method (Method)

These data visualization methods refer only to those used for patient communication. Although 9/19 clinicians interviewed explicitly mentioned using data visualization, only a very small variety of visualization techniques are referred to. **Line Graphs** and **Waffle Plots** are most common: line graphs for time-series data (e.g., blood pressure readings, disease trajectories or output from continuous glucose monitors); and waffle plots for probabilities of risks or side effects at a single point in time. **Color Coding** is also common, e.g., using traffic light systems to identify thresholds or numerical results e.g., as "normal" or "low risk"; "borderline, or medium risk"; and "abnormal or high risk". Finally, **Tree Diagrams** are occasionally used to visualize inheritance of disease causing genetic traits. This lack of diversity is also reflected in the patient information sheets analyzed with no additional methods identified.

5 APPLYING THE TYPOLOGY TO AID VISUALIZATION

When designing a data visualization, it is essential to define the data being visualized, the goal of the visualization and any relevant context. To illustrate how our typology aids in this, we now apply it to real world medical scenarios. We constructed these scenarios based on clinical vignettes from the medical literature [4] or clinician interviews, and verified the scenarios by two of the clinicians who participated in interviews.

EXAMPLE SCENARIO 1; DIABETES REVIEW APPOINTMENT: A patient with Type 2 diabetes attends the diabetes clinic for a *review* (intent: Evaluation and Explanation) of their glucose control, via a *haemoglobin A1c (HbA1c)* (feature: Indicator) *blood test* (context: Equipment and Procedures). The clinician discusses their recent results (Figure 1) and explains that despite presenting with good *blood glucose values* (type: Discrete Numerical Value) that day their *HbA1c* (feature: Indicator) has *increased* by 2.4% (Type: Discrete Numerical Value, Feature: Trend) from 7% to 9.4% (feature: Absolute Change). The clinician tells them this trend is extremely worrying, due to its sustained nature and puts them in the *high risk category* (feature: Threshold). They *urge them to better manage* (intent: Instruction) their diabetes by keeping their blood glucose within their *target range* (feature: Threshold) and avoiding *dangerous spikes* (feature: Outlier) in blood sugar levels.

In this scenario, our typology aids data visualisation by highlighting factors relevant to the method and presentation of a visualization. A suitable visualization must show that HbA1c blood tests are an indicator revealing long term trends in blood glucose levels, as opposed to day to day fluctuations. It should highlight how a small absolute change of 2.4% in HbA1c levels is an extremely significant relative change of 35% which predicts a higher probability of future complications. The visualization should be presented in the context of explaining to the patient why high HbA1c levels are problematic, relate them to their daily blood glucose levels. When advising on better diabetes control, physicians often refer to the underlying physiology of diabetes, and relate this to the data e.g., to

motivate the patient when instructing them to take insulin before a meal rather than after by showing graphs of blood glucose levels.

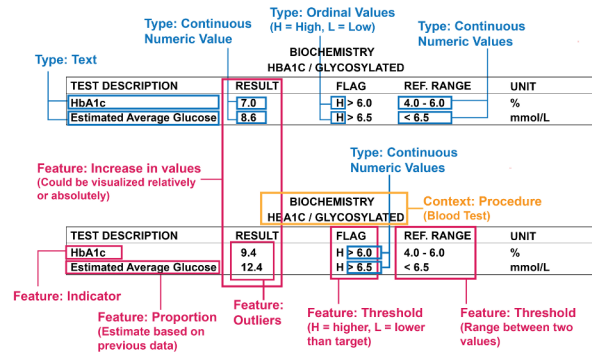


Figure 1: Typology applied to the table of blood test results discussed by the clinician and patient in Scenario 1. For clarity, not all instances of each classification are tagged. The annotation highlights how even in this seemingly simple example, medical data shows high complexity requiring careful consideration for visualization.

EXAMPLE SCENARIO 2; EMERGENCY CAESAREAN SECTION (C-SECTION): A woman presents at the maternity unit in labor. As labor is progressing slowly the clinician *discusses the option* (intent: Decision Support) of a *category 2* (type: ordinal value) emergency C-section and gives the patient a consent form to sign. *The form* (type: text) states that as the patient has previously had a C-section, there has been a *0.5% chance* (feature: Proportion) of scar rupture reported for those who proceed with vaginal delivery. They draw an *Anatomical Sketch* (context: Anatomy) to emphasize it is the internal scar that ruptures, not the external visible one.

In this scenario, visualizations must highlight that the risk is not a personalized probability for that patient, and instead a known proportion from existing research. To prevent misunderstanding, it is also important to indicate the anatomical location of the scar.

6 DISCUSSION

As the typology is constructed based on interviews with practicing clinicians, and analysis of current patient information sheets, it reflects the current state of the field. In doing so, it highlights missed opportunities, such as the lack of diversity in current visualization techniques (e.g., mostly waffle plots and line graphs) which can limit the depiction of more complex concepts (e.g., dependencies and interactions). It also explicitly highlights the hidden complexity of medical data which data visualizers must consider when designing communication aids. By recognizing these shortcomings and defining a vocabulary for data visualizers to engage with the medical community, we facilitate the exploration of clinical scenarios in which data visualization might aid clinician-patient communication. Additionally, we provide a common language to discuss necessary considerations to inform the design of successful medical data visualizations used in clinician-patient communication. In doing so, we hope to help address the growing problem of explaining and relating complex yet essential medical data to patients through the careful application of data visualization techniques.

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